

Modeling Charlotte Population Change versus Car Crashes

Overview

The regression model was used to analyze the relationship between two variables: population change and car crashes in Charlotte. We sought to answer the question: “For every person who moves to Charlotte, how many additional car crashes are expected?” Using regression, we aimed to determine if these variables move together and to establish a reliable framework for forecasting future counts.

Data and Features

We used SQL to merge two datasets to obtain both population and crash data for the Charlotte area covering the years 1975 to 2024.

Feature Set VIFs (Scores higher than 10 suggest multicollinearity):

log_population: 180.22

population_change: 17.05

population_growth_pct: 16.83

traffic_accidents_lag1: 6.68

traffic_accidents_growth_lag1:

1.38 year_trend: 539.44

population_roll3: 254.32

We decided to drop **log_population**, **year_trend**, and **population_roll3** because their extremely high Variance Inflation Factor (VIF) values indicated severe multicollinearity. We retained **population_change** and **population_growth_pct**. Although these scores are slightly above 10, they are significantly lower than the dropped features and are essential for estimating population impact within our model.

Model Building Process

We tested four different regression approaches to ensure we achieved the highest possible accuracy.

Linear Regression RMSE: 9,897.65

Random Forest RMSE: 10,192.00

KNN Regression RMSE: 10,225.25

GAM-style Splines RMSE: 10,821.24

Linear Regression was our best-performing model based on the RMSE test. It is a simpler model that uses fewer parameters. Utilizing a **TransformedTargetRegressor** allowed it to correctly evaluate the exponential nature of the real-world data, where we observed that variance increased as crashes grew. Our model was used to predict **log(Y)**, which aligned with our primary project goal.

Random Forest is a model that creates hundreds of decision trees and averages their results. It struggled with **overfitting**, as it produced a strong score with the training data but a much poorer holdout RMSE.

KNN Regression was not the best fit for our data because it struggles to predict outcomes outside of the training data range it has already seen. As population grows, KNN searches for the nearest neighbor values rather than following a mathematical formula. Since it was not trained on the higher population figures seen in later years, it predicted with less accuracy.

GAM-style Splines are useful when the crash-population relationship bends rather than staying linear. However, with a limited number of data points, the model had difficulty determining where to accurately place its curves.

Findings and Results

Based on our coefficients, **traffic_accidents_lag1** is our most influential variable. This indicates that a high percentage of the prediction can be attributed to the number of crashes reported in the immediate previous year.

The Bootstrap CI showed a 1.000 value, indicating 100% confidence in a positive correlation between population change and crashes. However, it showed only 95% confidence in a direct causal link. This is likely due to "noise" within our small dataset during years where crashes decreased unexpectedly, such as in 2020 during the COVID-19 pandemic, or due to shifts in safety patterns and traffic flow.

Limitations

Our dataset is small, consisting of only 42 observations and 9 test rows. This can cause high variance in estimates and coefficient instability, resulting in negative R^2 values and large coefficient swings. While the data helps interpret our project goal, the results must be used with caution.

Many similar population-related features suffered from multicollinearity and were pruned based on their VIF scores. The remaining predictors were useful for the model but still retained high VIF scores, which can inflate standard errors.

Conclusion

The primary objective of this project was to determine how population growth influences the total number of car crashes. Our analysis revealed that **Linear Regression with a log-transformed target** was the most effective model for this task.

While we initially expected modern machine learning techniques like Random Forest to perform better, the simplicity of the linear model allowed it to generalize more effectively across our 42-year dataset. The model successfully captured the exponential nature of urban growth, explaining approximately 83% to 87% of the variance during training.